

Name: _____ Date: _____

Definition 1. Recall the *product rule* (“first d-second plus second d-first”):

$$\frac{d}{dx} \left(f(x)g(x) \right) = f(x) \frac{dg}{dx} + g(x) \frac{df}{dx} \quad \text{or} \quad (f(x)g(x))' = f(x)g'(x) + g(x)f'(x)$$

Problem 1. For each of the following functions, use the product rule to find the derivative. Hint: you have to figure out what the functions $f(x)$ and $g(x)$ are. You may also have to use the product rule more than once!

a) $y = x \sin(x)$

b) $y = x^2 e^x$

c) $y = \sin(x) \cos(x)$

d) $y = 2e^x \tan(x)$

e) $y = (x^2 + 2x - 1) \ln(x)$

f) $y = x e^x \sin(x)$

g) $y = x^3 \tan(x) + x \tan(x)$

h) $y = x^3 \ln(x) \tan(x)$

$$i) \ y = \sqrt{x} \ln(x)$$

$$j) \ y = 2x^2 e^x \sqrt{x}$$

$$k) \ y = (x^4 - 3x^2 + 2) \ln(x)$$

$$\ell) \ y = \sin(x) \cos(x) \tan(x)$$

$$m) \ y = \sin^2(x)$$

$$n) \ y = \cos^2(x)$$

$$o) \ y = \ln^2(x)$$

$$p) \ y = e^{2x}$$